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AI Agents

Introduction





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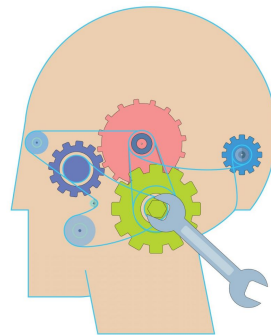
Why do we need Agents? AI Agents



From LLMs That Respond to Systems That Act

AI Agents

What happens when an AI doesn't just answer your question but has to actually accomplish a task?





A Task That Goes Beyond Chat

AI Agents

Imagine asking:

"Plan a 3-day trip to Dubai under \$1,500 and create an itinerary."

The system needs to:

- Understand your preferences
- Search for flights
- Compare options
- Search for hotels
- Check prices
- Build an itinerary
- Stay within budget



A Task That Goes Beyond Chat

AI Agents

THINK: Is generating one response enough?

NO

The task requires **multiple steps, decisions,**
and **external information.**



Understand
the question



Make
decisions



Retrieve external
information



Process and
reason



Generate final
response



What a Normal Chatbot Does?

AI Agents

A traditional chatbot typically follows:



<https://usamakhainights.medium.com/ai-chatbot-vs-ai-agent-whats-actually-the-difference-f86926dfe1fb>

For our travel example, it might say:

"Here are some flights and hotels you could consider."

But the user asked it to **complete a task**, not simply provide suggestions.



Limitations

AI Agents

The chatbot isn't necessarily deciding:

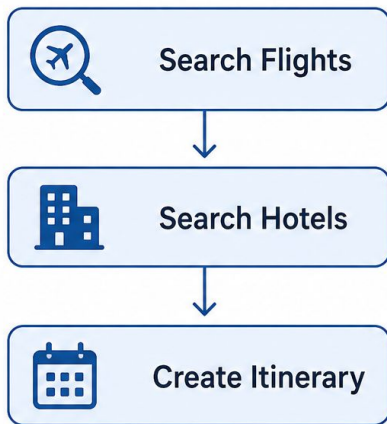
- What to do first
- Which tool to use
- What to do next
- Whether the goal has been achieved



A Fixed Workflow

AI Agents

We could manually define:



This works if the task always follows the same path.

But what if:

? No suitable flight is found?

? The hotel is over budget?

? The user changes their dates?

Now the system needs to **adapt**



The Need of AI Agents

AI Agents

For open-ended tasks, we want a system that can:

- Understand a goal
- Decide what to do
- Use tools
- Observe results
- Adjust its next step
- Repeat until the goal is achieved

This is where AI agents come in.



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How does AI Agents works?

AI Agents

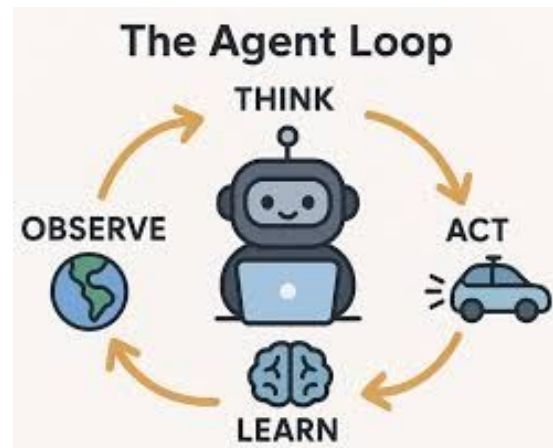


What Is an AI Agent?

AI Agents

An AI agent is a system that uses an **AI model** to work toward **a goal** by **deciding** what **actions** to take, **using available tools**, **observing results**, and **adapting its next steps**.

An agent doesn't just answer. It works toward an outcome.



<https://dev.to/sreeniS018/understanding-the-agent-loop-in-aws-strands-agent-framework-4hbn>



What makes a system Agentic?

AI Agents

An agent typically has:

Goal-directed behavior

Works toward a specific objective.

Reasoning

Interprets the situation and decides what to do.

Tool use

Interacts with external systems.

Iteration

Can perform multiple steps.

Adaptation

Changes its actions based on results.



Agent $\neq$ Just an LLM

AI Agents

An LLM can:

Understand $\rightarrow$ Reason $\rightarrow$ Generate

An agent adds:

Goal $\rightarrow$ Decision $\rightarrow$ Action $\rightarrow$ Feedback



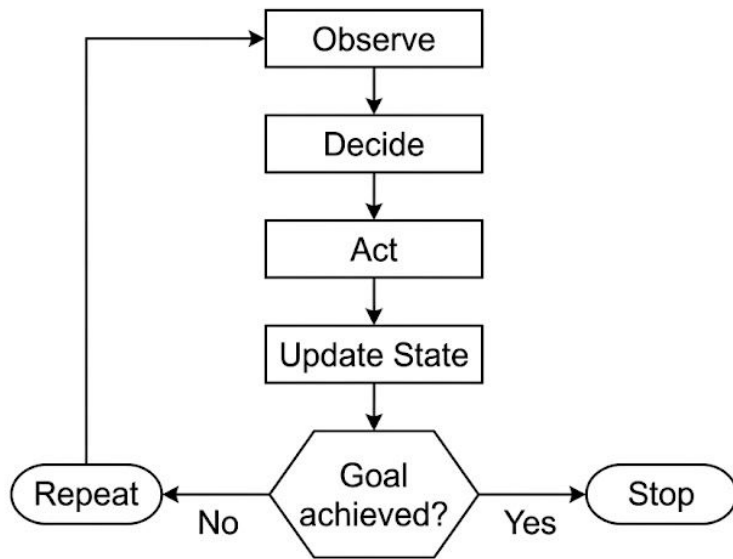
The LLM is an important part of an agent, but an agent is more than the LLM.



The Agent Execution Loop

AI Agents

An agent typically works through a loop:





Step1: Observe

AI Agents

The agent gathers the information it needs.

It may observe:

- User instructions
- Conversation context
- Memory
- Tool results
- Environment state

Travel Example

Goal:

"Plan my Dubai trip
under \$1,500."

→ The agent observes:





Step2: Decide

AI Agents

Goal:

The agent asks:
"What should I do
next?"

Agent determines subsequent
actions based on current results.

Task A: Flights



Use flight
search tool

Task B: Hotels



✓ Flight found



Use hotel
search tool

The next step can depend on the **current situation**.



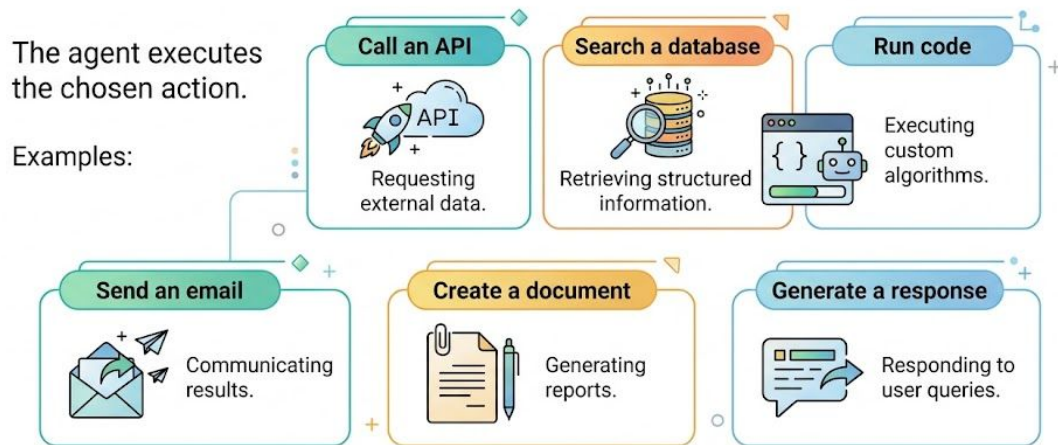
Step3: Act

AI Agents

The agent executes the chosen action.

Examples:

- Call an API
- Search a database
- Run code
- Send an email
- Create a document
- Generate a response



The engineer didn't relearn programming.



Step4: Update State

AI Agents

The agent records what happened.

For example:

Budget: \$1,500

Flight: \$600

Remaining: \$900

Hotel search: Needed

Now the next decision has more context.

Why?

Without state, the agent may not know:

What has already happened?



Step5: Stop or Repeat

AI Agents

After each action:

Has the goal been achieved?

If yes → stop

If no → continue

Action



Result



Goal achieved?

└─ Yes → Finish

└─ No → Decide next action

Why?

Without stopping conditions, an agent can:

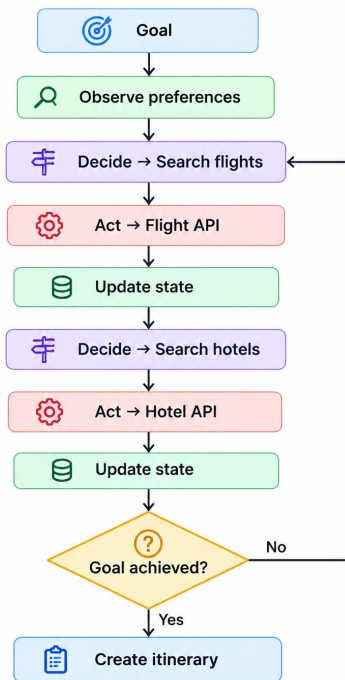
- Keep calling tools
- Waste money
- Repeat actions
- Get stuck in loops



The Complete Loop

AI Agents

Let's return to our travel task.



This is the core mental model of an agent.



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Components AI Agents



What does an Agent Need?

AI Agents

A practical agent system needs several components:

1. **LLM**
2. **Planning / Orchestration**
3. **Tools**
4. **Memory / State**
5. **Evaluation & Guardrails**



LLM: The Reasoning Engine

AI Agents

The LLM interprets the goal and helps determine the next action.

It can:

- Understand instructions
- Interpret context
- Reason about the task
- Select tools
- Generate decisions or responses



<https://medium.com/@xsankalp13/llm-vs-agentic-ai-understanding-the-evolution-of-artificial-intelligence-1c86e107765a>

LLM = Brain



Model Choice Matters

AI Agents

Different models have different trade-offs.

Consider:

- **Cost** — How expensive is each request?
- **Speed** — How quickly does it respond?
- **Reasoning quality** — How well does it handle complex tasks?
- **Context capability** — How much information can it handle?
- **Task requirements** — What does your agent actually need?

There is no single "best" model for every agent.



Planning & Orchestration

AI Agents

Planning determines:

"What should the agent do next?"

It may:

- Break a goal into smaller tasks
- Decide task order
- Choose the next action
- Coordinate multiple tools or agents

Research AI agents and write a report.





Tools: The Agent's Hands

AI Agents

Tools give the agent capabilities beyond generating text.

Examples:

- APIs
- Databases
- Search
- Code execution
- External applications

Tools turn reasoning → execution.



Memory & State

AI Agents

Memory and state preserve information needed during and across tasks.

Short-term memory

Current conversation, task context, and recent tool results.

Long-term memory

Persistent information such as preferences and recurring instructions.

Experience memory

Past actions, outcomes, and lessons.



Memory vs State

AI Agents

Keep this distinction simple:



State

What is happening right now?

Example: Flight search completed → hotel search pending.



Memory

What should the system remember?

Example: User prefers window seats.



Together

They help the agent maintain **continuity across steps**.



Evaluation & Guardrails

AI Agents

An agent needs mechanisms that answer two questions:

Evaluation

"Is the agent doing a good job?"

Monitor:

- Task success
- Errors
- Tool usage
- Output quality

Guardrails

"Is the agent staying within safe limits?"

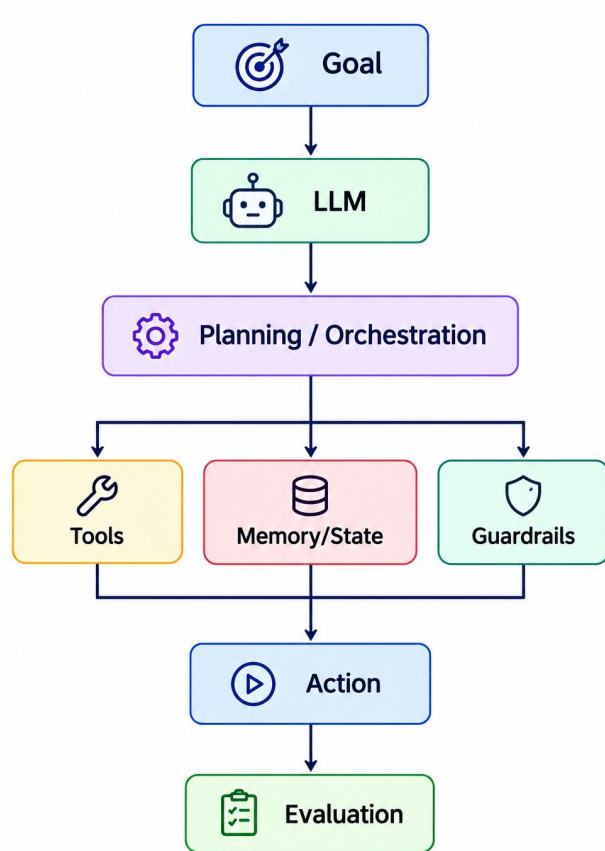
Control:

- What tools it can access
- What actions it can perform
- What information it can use
- When human approval is required



The Agent System

AI Agents





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Agents, Chatbots, and Workflows

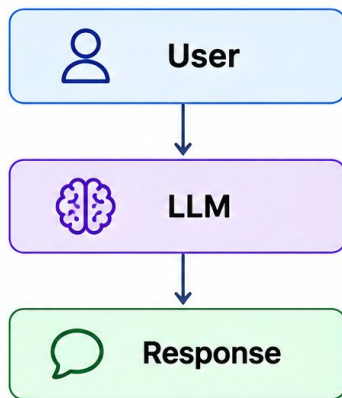
AI Agents



Chatbot vs Agent

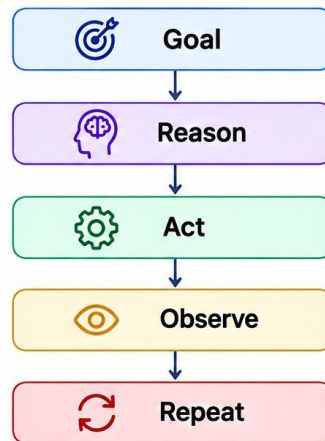
AI Agents

Chatbot



Primarily responds to user input.

Agent



Works toward a goal using multiple steps.

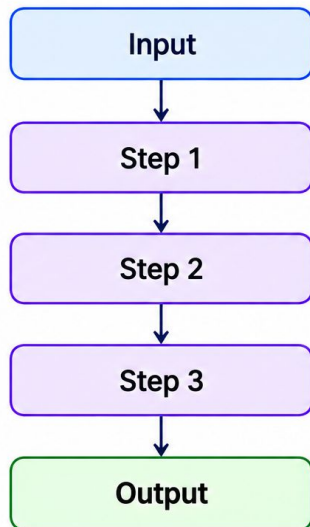


What Is a Workflow?

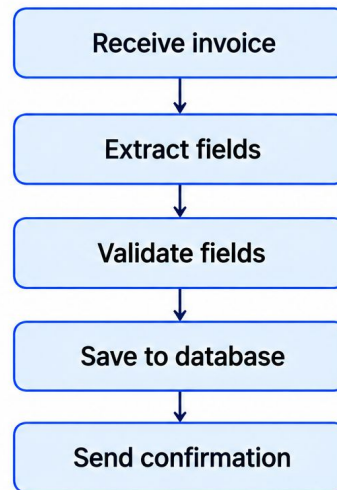
AI Agents

A workflow is a system where **humans define the sequence of steps in advance**.

General Workflow



Example





When Should We Use a Workflow?

AI Agents

Use a workflow when the task is:

- Repetitive
- Predictable
- High-volume
- Well-defined
- Easy to represent as fixed steps

Examples

- ETL pipelines
- Report generation
- Batch processing
- Fixed approval processes



When Should We Use an Agent?

AI Agents

Use an agent when the task is:

- Ambiguous
- Open-ended
- Multi-step
- Dynamic
- Context-dependent
- Requires planning

Examples

- Research assistant
- Personal assistant
- Complex task automation
- Business analysis



Workflow vs Agent

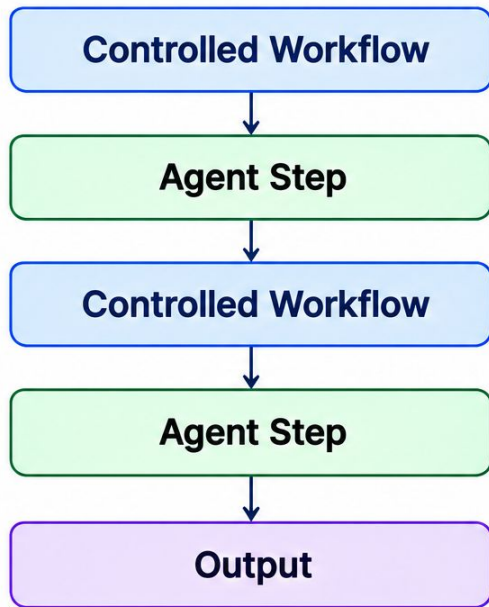
AI Agents

	Workflow	Agent
Steps	Predefined	Dynamically selected
Best for	Predictable Tasks	Open-ended Task
Control	High	More Flexible
Adaptability	Limited	High
Debugging	Easier	More Challenging
Runtime Decision	Few	Many



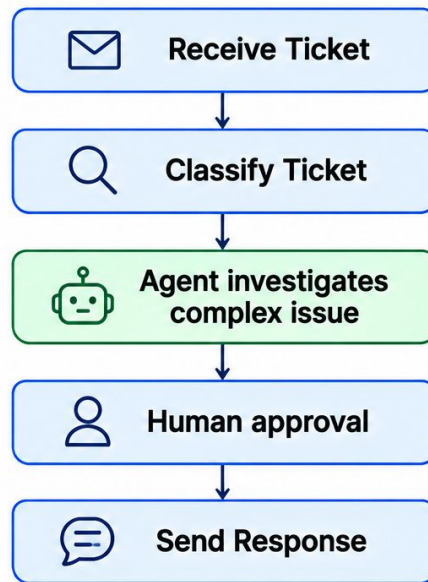
Why Hybrid Systems Are Common

AI Agents



Example

A customer-support workflow might be:



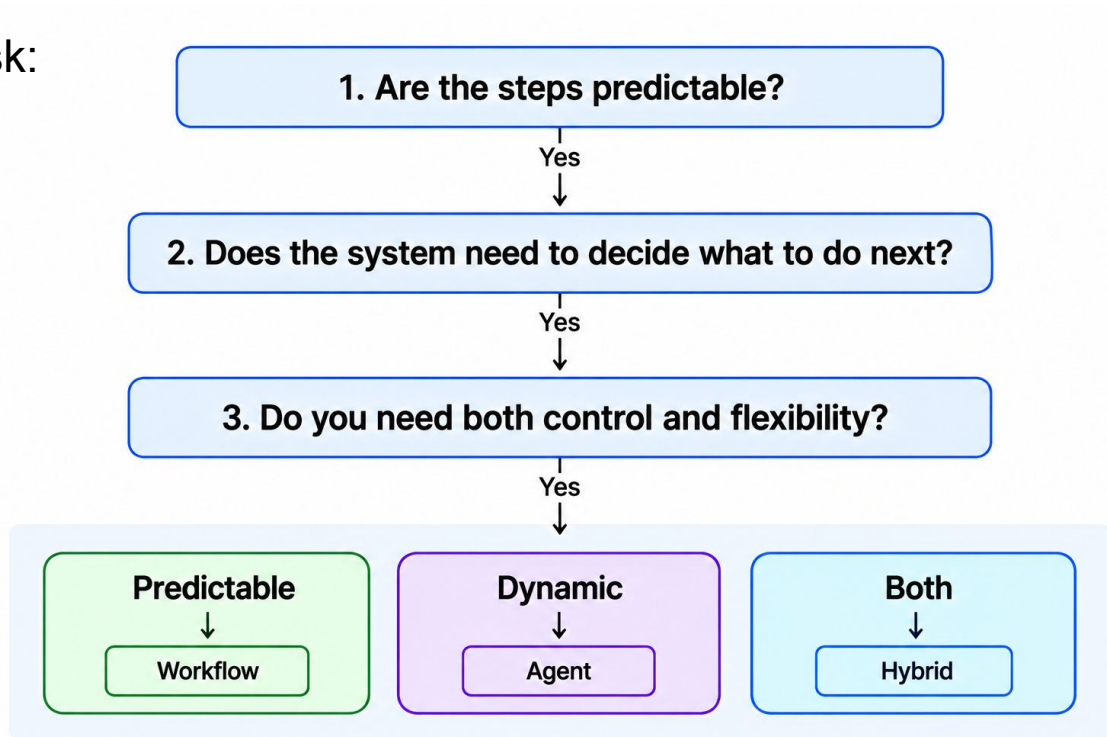
Agents are often used inside controlled workflows—not necessarily instead of workflows.



Decision Guide

AI Agents

Ask:





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Autonomy AI Agents



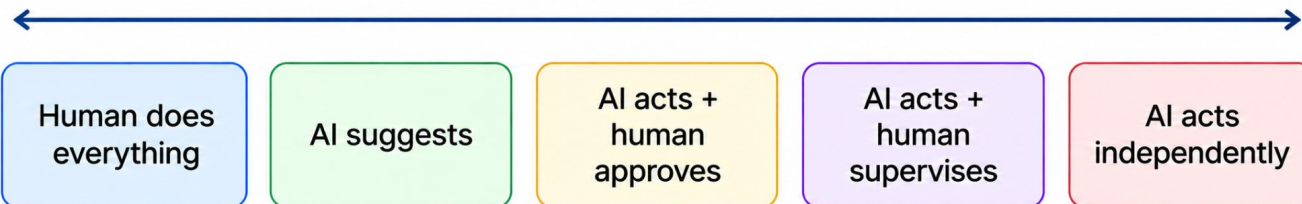
What Is Autonomy?

AI Agents

Autonomy means:

How independently can the system act without human intervention?

Think of a spectrum:



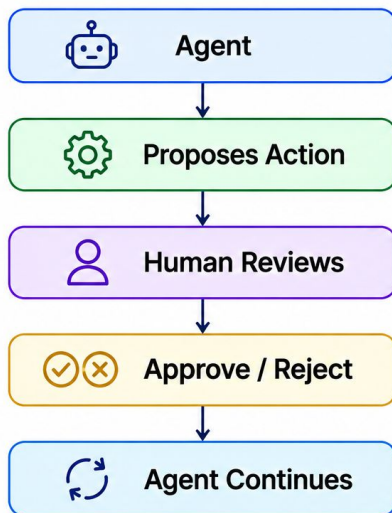


Human in the Loop

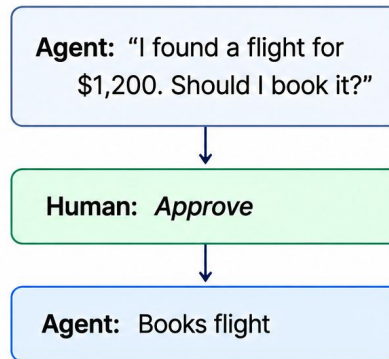
AI Agents

For important actions, the agent can ask for approval.

Agent-Human Approval Loop



Example





Permission and Guardrails

AI Agents

Permissions

What can the agent access?

Guardrails

What is the agent allowed to do?

Example

- Read-only access to some systems
- Spending limits
- Approved tools only
- Restricted data access
- Human approval for high-risk actions



Stopping Conditions

AI Agents

An agent needs clear limits.

Examples:

- Maximum number of tool calls
- Maximum spending
- Maximum execution time
- Stop after repeated failures
- Stop when human approval is required

More autonomy does not mean unlimited autonomy.



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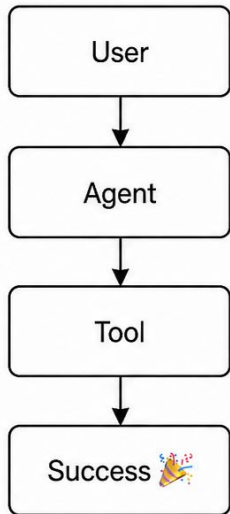
From Demo to Production AI Agents



Why Agent Demos Fail in Production

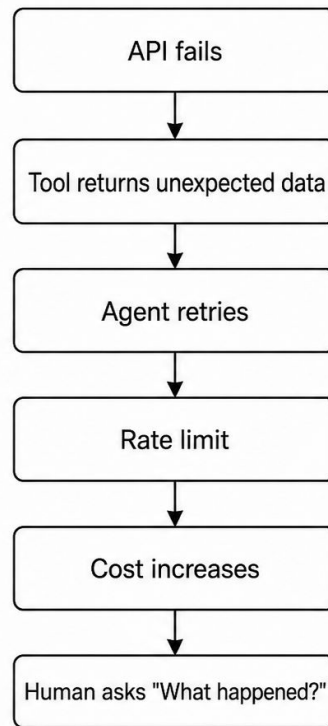
AI Agents

Demo



A successful demo does not automatically mean a reliable production system.

Production





What Production Agents Need?

AI Agents

Reliability

Handle:

- Tool failures
- API errors
- Unexpected outputs
- Retries

Observability

Track:

- What the agent did
- Which tools it called
- How long it took
- What went wrong

Cost & Performance

Monitor:

- Token usage
- Number of tool calls
- Latency
- Cost per run



How Do We Know the Agent Worked?

AI Agents

❌ Don't only ask:

"Was the final response correct?"

✅ Ask instead:

"Did the agent actually complete the task?"

🎯 Goal:

✈️ Book a flight.

⚠️ A response saying:

"Flight successfully booked."
is **not enough**.



We need to verify: →



Search Flights



Select Flight



Book Flight



Booking Confirmation





Safety Risks

AI Agents

Agents can interact with real systems, so failures can have consequences.

Common risks:

- **Prompt injection** — malicious instructions influence the agent
- **Tool misuse** — unsafe actions are performed
- **Data leakage** — sensitive information is exposed
- **Wrong objective** — the agent optimizes for the wrong outcome



Basic Controls

AI Agents

Guardrails

Filter and constrain inputs/outputs.

Permissions

Limit tool and data access.

Human approval

Require confirmation for sensitive actions.

Sandboxing

Isolate potentially risky actions.



Summary

- **AI Agents:** LLM-powered systems that can **reason, use tools, and take actions** to achieve a goal.
- **How they work:** **Observe** → **Decide** → **Act** → **Update State** → **Repeat/Stop**
- **Core components:** **LLM + Planning + Tools + Memory/State + Guardrails & Evaluation**
- **When to use:** Agents for **dynamic, multi-step tasks**; workflows for **predictable tasks**; hybrid systems can combine both.
- **Control:** Greater autonomy requires **permissions, guardrails, human approval, and stopping conditions**.
- **Production:** Reliable agents need **observability, error handling, evaluation, and safety controls**.



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Thank You

We'd love to hear from you. If you have any questions,
feel free to reach out to our team anytime!